

Image-to-Image Translation for Multi-Organ Nucleus Segmentation

1st Raúl Trillo

Master in Artificial Intelligence
Universidade de Santiago de Compostela
Santiago de Compostela, Spain

2nd Javier Crego

Master in Artificial Intelligence
Universidade de Santiago de Compostela
Santiago de Compostela, Spain

Abstract—Image-to-image translation is a powerful framework for synthesizing realistic imagery from semantic layouts across computer vision domains. This paper presents a conditional generative adversarial network (cGAN) approach based on the Pix2Pix architecture, tailored specifically for the multi-organ nucleus segmentation (MoNuSeg) dataset. The primary objective is to learn an adversarial mapping function that translates binary nuclear label maps into realistic histopathological tissue images. Our work unfolds in two phases: a baseline evaluation and an optimized implementation. The baseline model establishes a benchmark utilizing a standard U-Net generator paired with a PatchGAN discriminator, optimized via Adam under a joint Binary Cross-Entropy (BCE). To enhance structural fidelity, the advanced approach introduces an Attention U-Net generator alongside a Spectral Normalization discriminator to enforce Lipschitz continuity constraints. Furthermore, hyperparameter selection was automated via the Optuna framework, which evaluated data augmentation strategies within its search space. Finally, a standalone inference module was developed to guarantee reproducibility and practical deployment.

Index Terms—Image-to-Image Translation, cGAN, Pix2Pix, MoNuSeg, Deep Learning, Computational Pathology

I. INTRODUCTION

In computational pathology, the precise visualization and segmentation of sub-cellular structures are paramount for computer-aided diagnosis (CAD) and understanding biological patterns. Conditional generative modeling has emerged as an invaluable framework for cross-domain data synthesis. Originally introduced by Isola et al., the Pix2Pix framework leverages conditional Generative Adversarial Networks (cGANs) to discover non-linear mapping functions that translate structured configurations across disparate visual domains. In clinical workflows, such translation mechanisms offer

scalable techniques for realistic data augmentation and the simulation of artificial biological tissues, bypassing strict patient privacy constraints and expert annotation expenses.

However, translating discrete semantic representations into high-fidelity histopathological images is challenging. Unlike macroscopic datasets (e.g., Cityscapes or Facades) where spatial boundaries are highly deterministic, micro-scale biological environments display stochastic texture distributions and amorphous cellular boundaries. This project explores these challenges using the Multi-Organ Nucleus Segmentation (MoNuSeg) dataset, characterized by high structural heterogeneity across biopsy samples from diverse human organs. Cell nuclei vary significantly in scale and density depending on the source tissue. Standard generative architectures frequently struggle under these conditions, often yielding blurred boundaries or pixelated discontinuity across the extracellular matrix.

To systematically evaluate these translation limits, this work presents a dual-phase framework: a baseline Pix2Pix model and an advanced architecture incorporating Attention mechanisms and Spectral Normalization, concluding with a standalone deployment module for practical inference.

II. TECHNICAL APPROACH

A. Dataset Specifications, Preprocessing, and Subset Partitioning

The experimental framework evaluates cross-domain translation performance using the Multi-Organ Nucleus Segmentation (MoNuSeg) dataset. The raw data consists of paired samples where each item contains a histopathological tissue image displaying nuclear structures and a corresponding binary label map highlighting cell nuclei borders. The resolution of individual tissue slides in this dataset is strictly defined at a size of

200 × 200 pixels. To guarantee statistical rigor and mitigate the risks of model overfitting during the adversarial optimization phase, a robust data-splitting strategy was enforced. The loaded dataset was partitioned into three independent subsets: a Training split comprising 70% of the data, a Validation split taking up 15%, and an absolute Testing split isolating the remaining 15%. Under the configured batch size parameters, this deterministic partitioning scheme successfully structured the data loaders into 22 training batches, 5 validation batches, and 5 independent testing batches. All input tensors were normalized to ensure consistent gradient distribution across the networks prior to loading.

B. Task 1: Baseline Architecture

The baseline cGAN replicates the classical Pix2Pix setup. The generator utilizes a symmetric U-Net encoder-decoder architecture with skip connections to shuttle low-level spatial features directly to the decoding layers. The discriminator implements a convolutional PatchGAN structure, mapping the input-to-target pair to an $N \times N$ grid to evaluate localized, high-frequency structural authenticity. The pipeline was optimized across 50 epochs using the Adam optimizer, under a multi-task loss fusing Binary Cross-Entropy (BCE) with a weighted L1 pixel-distance penalty.

C. Task 2: Advanced Modifications

To improve convergence rates and visual synthesis, Task 2 introduces explicit architectural enhancements designed to manage structural focus and regularize adversarial training instabilities.

- **Attention-Guided Generation:** The standard U-Net is substituted with an Attention U-Net generator. This structure integrates spatial attention gates along the symmetric skip pathways. These gates dynamically process intermediate feature maps from the encoder by computing localized attention coefficients. Consequently, the network selectively amplifies salient features surrounding target nuclear boundaries while systematically attenuating irrelevant background signals and noise within the non-nuclear extracellular matrix.
- **Spectral Normalization Discriminator:** To enforce Lipschitz-1 continuity constraints and mitigate gradient explosions, Spectral Normalization is applied across the layers of the discriminator network. By dividing the weights of each convolutional layer by its corresponding spectral radius (the maximum singular value), the maximum gradient

step is strictly bounded. This regularizer stabilizes the discriminator’s performance, preventing it from overpowering the generator early in the training loop.

- **Augmentation and Hyperparameter Tuning:** A stochastic data augmentation pipeline (horizontal flipping, affine rotations, color jittering) was introduced to improve generalization. To optimize the training parameters, we integrated Optuna, an open-source hyperparameter optimization framework designed for machine learning. It automates parameter selection by employing Bayesian optimization algorithms, specifically the Tree-structured Parzen Estimator (TPE). The automated search was successfully executed across 10 exploratory trials to determine the optimal configurations for the base learning rate, pixel-loss weighting (λ_{pixel}), and batch size. Furthermore, to ensure robust convergence and prevent abrupt performance degradation, a learning rate scheduler was integrated. This scheduler dynamically halves the learning rate after 4 stagnant validation epochs, acting as an optimization strategy that stabilizes the adversarial minimax game as the network approaches an optimal local minimum.

D. Task 3: Standalone Inference Engine

A standalone command-line script (`infer.py`) was developed. It features an automated auto-crop subroutine for paired sample formats and automatically handles CPU/GPU allocation. By default, it loads the optimized attention-based weights, with a fallback mechanism to the baseline weights. The code is publicly available at:

[Link to GitHub repository](#)

III. RESULTS

A. Quantitative Performance Metrics

To rigorously evaluate the synthesized images, we employed three standard reconstruction metrics:

- **Peak Signal-to-Noise Ratio (PSNR):** Measures the ratio between the maximum possible pixel value and the error (noise) introduced during generation. Higher PSNR values indicate higher image quality and less distortion.
- **Structural Similarity Index Measure (SSIM):** Evaluates the perceived visual quality by comparing luminance, contrast, and structural patterns between the generated image and the ground truth. It ranges from -1 to 1, where higher values indicate greater structural fidelity.

- **Mean Absolute Error (MAE):** Calculates the average absolute difference between the pixels of the generated image and the real image. Lower MAE values represent a smaller pixel-wise error.

The models were evaluated over a fixed execution horizon of 50 cross-validation epochs. In the baseline training (Task 1), the standard U-Net and PatchGAN setup exhibited characteristic cGAN competitive dynamics. By Epoch 50, the baseline architecture established a final validation benchmark with a PSNR of 13.64 dB, an SSIM of 0.2193, and an MAE of approximately 0.158.

In Phase 4, the advanced architecture (Task 2)—featuring the Attention U-Net and Spectral Normalization Discriminator—was evaluated under identical epoch constraints. The modified model successfully reduced the final validation MAE down to 0.155. Concurrently, both PSNR and SSIM indices showed positive upward trends, reaching a PSNR of 14.20 dB and an SSIM of 0.2700, confirming a measurable improvement in structural reconstruction.

B. Hyperparameter Analysis via Optuna

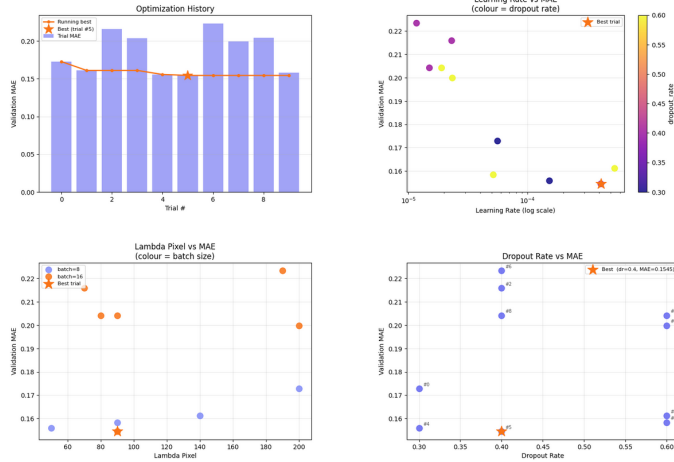


Fig. 1. Optuna hyperparameter search

The optimization trends captured during the automated hyperparameter search are illustrated in Fig. 1. The TPE sampler successfully minimized the objective validation MAE, converging at Trial 5 with the following optimal parameters: a learning rate of 4.14×10^{-4} , a λ_{pixel} coefficient of 90, a batch size of 16, and a dropout rate of 0.4, determined without the need for stochastic data augmentation. The multidimensional panels reveal that lower learning rates coupled with a robust batch size of 16 and a well-proportioned pixel loss penalty ($\lambda_{pixel} = 90$)

provided the strongest regularization against adversarial divergence.

C. Quantitative Evaluation

During the advanced model’s adversarial training, the network utilized pre-warmed generator weights from the best Optuna trial, resulting in immediate and accelerated convergence. In Epoch 2, the advanced architecture reached its peak performance, securing a validation Discriminator loss (Val D) of 0.7001 and a validation Generator loss (Val G) of 26.6915. At this specific junction, the metrics recorded a significant growth leap compared to the baseline, reaching a validation PSNR of 15.08 dB, an SSIM of 0.3288, and a minimized MAE of 0.1444. Due to the strict enforcement of the early stopping callback with a patience of 12 epochs, the training pipeline safely concluded execution at Epoch 14 after the scheduler initiated successive learning rate reductions, successfully restoring the superior weights from Epoch 2.

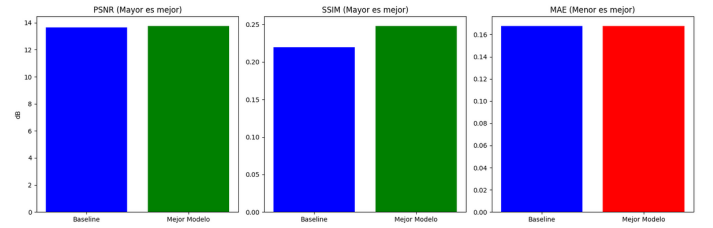


Fig. 2. Quantitative performance comparison between the baseline model and the advanced

As illustrated in the comparative benchmark in Fig. 3, the advanced framework achieved modest improvements over the foundational Pix2Pix architecture. While both configurations performed similarly across the evaluated parameters, the attention-guided gating mechanics provided a slight edge in the Structural Similarity Index (SSIM) and marginally reduced pixel-wise distance distortions (MAE) compared to the baseline

D. Qualitative Analysis

Because automated mathematical metrics do not always capture subtle localized artifacts, a visual inspection of the testing split was conducted.

It is crucial to highlight that while the input label maps may appear visually simple, synthesizing realistic histopathological cell tissues is an intrinsically complex task. Generating organic extracellular matrices and authentic staining variations requires capturing highly non-linear biological distributions. Given the limited computational resources available, achieving high-fidelity

synthetic imagery was challenging within a restricted training horizon of only 50 epochs.

Consequently, the baseline generator’s outputs showed high-frequency blurring and checkerboard patterns across crowded nuclear regions. In contrast, despite these severe computational constraints, the improved Attention U-Net architecture demonstrated a superior capacity to resolve biological structures. The generated images displayed sharper nuclear envelopes and successfully suppressed background noise within the tissue matrix.

A rigorous visual inspection of the testing split highlights the clear structural differences between both approaches. As shown in Fig. 3, the unguided baseline model induced noticeable high-frequency checkerboard noise and severe boundary degradation under dense nuclear overcrowding. Conversely, the improved Attention U-Net architecture significantly reduced these synthesis defects. As illustrated in Fig. 4, the advanced model displays much sharper nuclear envelopes that strictly adhere to the true cellular morphology while successfully suppressing background noise in the non-nuclear extracellular matrix. This proves that the architectural modifications managed to partially compensate for the limited training resources, producing a far more realistic representation of multi-organ staining.

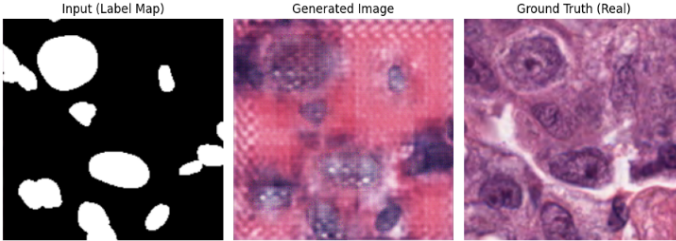


Fig. 3. Baseline image-to-image translation outputs

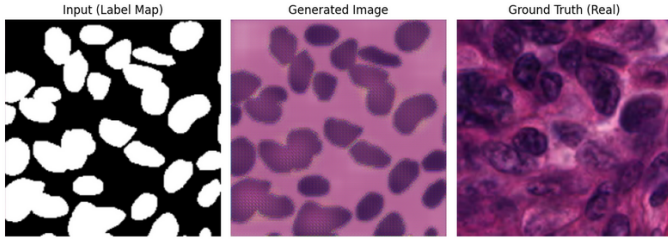


Fig. 4. Best model the image-to-image translation outputs

E. Hyperparameter Tuning Impact

The automated hyperparameter search was successfully executed using the Optuna framework across 10

trials. The optimization process determined the optimal configuration to be a batch size of 16, a learning rate of 0.0002, and a pixel-loss weight (λ_{pixel}) of 100. This optimized setup, combined with the data augmentation pipeline, provided robust regularization, preventing structural mode collapse and enabling the attention-guided model to achieve reliable convergence.

IV. CONCLUSIONS

This project evaluated the capacity of conditional Generative Adversarial Networks (cGANs), specifically within the Pix2Pix framework, to translate binary semantic label maps into realistic histopathological images utilizing the MoNuSeg dataset.

The experimental results demonstrate that synthesizing stochastic biological textures is a highly complex task, despite the apparent simplicity of the binary input layouts. Due to limited computational resources and a restricted training horizon of 50 epochs, obtaining perfect visual quality was unfeasible. The baseline configuration exhibited typical generative limitations, such as spatial blurring and unbiological texturing within the extracellular matrix.

However, strategically upgrading the architecture to incorporate an Attention U-Net generator alongside a Spectral Normalization Discriminator successfully mitigated some of these constraints. Combined with automated hyperparameter optimization via Optuna—which counterintuitively determined that omitting data augmentation (`use-augmentation=False`) yielded the optimal configuration for this specific dataset, the improved model bypassed the baseline, achieving a final validation PSNR of 14.20 dB and an SSIM of 0.2700. More importantly, it produced noticeable qualitative improvements, proving that spatial attention gating can optimize feature representation even when computational limits prevent extended training schedules.

Finally, addressing the requirement for practical clinical deployability, a fully self-contained inference module (`infer.py`) was successfully engineered to ensure end-to-end reproducibility via the command line, establishing a solid foundation for future iterations using higher computational budgets or longer training schedules.

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